

Qixuan Min

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RESEARCH INTERESTS

Metasurface, Computational Imaging, Photonics Chip, Structured Light.

EXPERIENCE

Shanghai Institute of Optics and Fine Mechanics, CAS, Shanghai, China Research Assistant & Associate Engineer	July 2024 — June 2026
University of Chinese Academy of Sciences, SIOM, Shanghai, China Master of Science	Sep 2021 — July 2024 GPA: 3.7/4.0
Thesis Title: The Study of Quantitative Phase Imaging Based on Polarization-Multiplexed Metasurface Thesis Advisor: Prof. Guohai Situ	Mark: 95/100
Shanghai Maritime University, Shanghai, China Bachelor of Automation(Outstanding Graduates)	Sep 2017 — July 2021 GPA: 3.3/4.0 Rank(Top5.6%)

PUBLICATIONS

Journal paper

- [1] **Min, Q.**, Trapp, J., Fang, T., Hu, R., Wang, F., Zhang, Z., Liu, X., Dai, A., Yang, C., Guo, J. and Situ, G., 2024. Varifocal Metalens for Compact and Accurate Quantitative Phase Imaging. **ACS Photonics**, 11(7), 2797-2804.
- [2] Hu, R., **Min, Q.**, Liu, X., Dai, A., Guo, J. and Situ, G. 2024. Terahertz programmable metasurface for phase modulation based on free carrier plasma dispersion effect. **Applied Physics Letters**, 124(25), 251703.
- [3] Dai, A., Fang, P., Gao, J., **Min, Q.**, Hu, R., Qiu, S., Wu, X., Guo, J. and Situ, G., 2023. Multifunctional Metasurfaces Enabled by Multifold Geometric Phase Interference. **Nano Letters**, 23(11), 5019-5026.
- [4] Zhang, Z., Wang, F., **Min, Q.**, Jin, Y. and Situ, G., 2024. Fourier phase retrieval using physics-enhanced deep learning, **Optics Letters**, 49(21), 6129-6132.
- [5] Zhang, N., Wang, F., **Min, Q.**, Liu, X., Yuan, H., Guo, J. and Situ, G., 2025. Broadband and polarization-independent complex amplitude modulation using a single layer dielectric metasurface. **Nanoscale**, 17(15), pp.9562-9568.
- [6] Zhang, N., **Min, Q.**, Liu, X., Yuan, H., Liu, H., Huang, Z., Zhao, X., Guo, J. and Situ, G., 2025. SiM-SHDR: Single-Layer Metasurface for Snapshot High Dynamic Range Imaging, **Photonics Research**, 13 (8), 2270-2280

Conference

- [1] **Min, Q.**, Guo, J. and Situ, G. 2024 Compact Quantitative Phase Imaging Based on a Polarization-Dependent Varifocal Metalens. **Optica Digital Holography and Three-Dimensional Imaging. 2024. Italy (Oral Presentation)**

AWARDS

Monash Graduate Scholarship This scholarship covers tuition and living costs for a PhD at Monash.	Melbourne, Australia May 2025
Entrepreneurial Pioneer Award The award is given to those teachers and students who are brave in entrepreneurship.	Hangzhou, China Nov 2021
Outstanding Freshmen Scholarship The award is given to the outstanding new students of the University of the Chinese Academy of Sciences.	Hangzhou, China Sep 2021
Outstanding Graduates The award is given to the most outstanding undergraduate graduates, equivalent to a first-class honors degree.	Shanghai, China June 2021
First Price in TI Cup Undergraduate Electronics Design Contest A nationwide undergraduate student competition with high prestige, featuring over 20,000 participants and less than 300	Shanghai, China

first-place winners, a ratio of 2%.

Second Price in NXP Cup National Undergraduate Smart Car Contest

Sep 2019
Shanghai, China

A nationwide undergraduate student competition with high prestige, where only a very few teams can advance to the finals and win national awards.

Third Price in NUETIAC

Aug 2019
Shanghai, China

A nationwide undergraduate student competition with high prestige, where only a very few teams can advance to the finals and win national awards.

May 2019

RESEARCH EXPERIENCE

Broadband, large field-of-view Achromatic Imaging using Hybrid Meta-Optics

Sep 2025 — Present

This project aims to design a broadband achromatic refractive-meta hybrid lens with a wide field-of-view using ray tracing, leveraging 3D printing for direct integration onto imaging devices

- Proposed a scheme for a wide-FoV, achromatic refractive-meta hybrid lens.
- Designed a single-let refractive-meta hybrid lens.
- Performed achromatic characterization experiments.
- Performed achromatic imaging experiments.

Polarization-dependent varifocal metalens for quantitative phase imaging

Sep 2022 — June 2024

The aim of this project is to design a highly integrated and high-precision quantitative phase imaging device based on a polarization-dependent varifocal metalens.

- Proposed a compact quantitative phase imaging method using polarization-dependent varifocal metalens.
- Designed and simulated the varifocal metalens.
- Performed quantitative phase imaging experiments and wrote the manuscript.
- Made an oral presentation in Optica DH2024.

Terahertz programmable metasurface

Dec 2021 — Jan 2024

The goal of this project is to design a high-speed, high-efficiency reconfigurable metasurface based on an MIM structure.

- Carried out simulations of the PN junction and analyzed the data.

Multifold geometric phase metasurfaces based on interference effect

Sep 2021 — Aug 2022

The aim of this project is to design a complex amplitude modulation geometric phase metasurface using the interference coupling effect between multiple meta-atoms.

- Conducted experiments on spin-decoupled metalens imaging.

SKILLS

- **Programming:** Python, Matlab, C, C++
- **Software:** Lumerical, CST, COMSOL, Altium Designer, Solidworks, 3Ds Max, Fusion 360, Git and so on.

REFERENCES

Prof. Haoran Ren

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Prof. Guohai Situ

Professor, Director at Shanghai Institute of Laser Technology, Shanghai, China

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Prof. Shensheng Han

Professor, Shanghai Institute of Optics and Fine Mechanics, Shanghai, China

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